

Total No. of Questions—8]

[Total No. of Printed Pages—3

Seat No.	
-------------	--

[4657]-576

S.E. (Computer) (Second Semester) EXAMINATION, 2014

OBJECT ORIENTED AND MULTICORE PROGRAMMING

(2012 PATTERN)

Time : Two Hours

Maximum Marks : 50

N.B. :— (i) Attempt Q. No. 1 or Q. No. 2, Q. No. 3 or Q. No. 4,
Q. No. 5 or Q. No. 6, Q. No. 7 or Q. No. 8.

(ii) Neat diagrams must be drawn wherever necessary.

(iii) Figures to the right indicate full marks.

1. (a) What is a friend function ? How do you declare one ? When is friend function compulsory ? Give an example. [6]
- (b) What is virtual function ? Why do we need virtual function ? When do we make a virtual function “pure” ? What are the implications of making a function a pure virtual function ? [6]

Or

2. (a) Write short notes on : [6]
- (a) ‘this’ pointer
- (b) Copy constructor.

P.T.O.

- (b) What is operator overloading ? Name the operators that cannot be overloaded in C++ ? How do you declare an overloaded stream insertion and extraction operator ? [6]
3. (a) Distinguish between overloaded functions and function templates. Write a function template for finding the minimum value contained in an array. [6]
- (b) Draw the state diagram for the process. Explain each process state briefly. [6]

Or

4. (a) What is an exception ? How is an exception handled in C++ ? What are the advantages of using exception handling mechanism in program ? [6]
- (b) How can we create a child process from parent process ? Describe parent-child relation. [6]
5. (a) What do you mean by threads ? Write a threaded program in C++ and explain Pthread_join() function. [7]
- (b) Compare threads and processes. Describe thread attributes. [6]

Or

6. (a) How can we pass command line arguments to the thread function ? Determine the number of threads using command line argument. [7]
- (b) What are the types of thread ? Describe in brief. [6]

7. (a) What is meant by critical section ? How can we manage critical section using mutex semaphore ? [6]

(b) Enlist concurrency models. Write a short note on any *two*. [7]

Or

8. (a) Can all threads in the process share global data, variables, and data structure ? If yes, explain with an example. [6]

(b) Describe the basic functions of POSIX message queue. [7]